Seema Rida

San Diego, CA seemazrida@gmail.com (858)900-7565 seemazr.com seemarida
seemarida

Summary

Cognitive Science student specializing in Machine Learning & Neural Computation with experience in experimental design, cognitive and behavioral research, and human subjects research. Strong technical foundation in Python, machine learning, data science, and signal processing, with expertise in building data pipelines, analyzing multimodal signals, and performing advanced analyses across educational technology, brain-computer interfaces, human development, and computational neuroscience.

Skills

Programming: Python (pandas, scikit-learn, NumPy, matplotlib, PyTorch), Git/GitHub, Java, C, LaTeX, Vim
Data Science & ML: Machine Learning (classification, clustering, predictive modeling), Feature Engineering, PCA, K-means, RNNs, Data Visualization (Matplotlib, Seaborn, Plotly) Signal Processing and Systems: Biometric Analysis, Neural Signal Processing, Sensor Data Parsing, Lab Streaming Layer (LSL), Embedded systems, network configuration, managed switches, PTP/Dante networking
Research: CITI-certified, Experimental Design, Human-Subjects Research, Multimodal Data Collection, Behavioral Coding, Scientific Writing & Presentations

Education

University of California San Diego *Sept 2022 – June 2026*
BS in Cognitive Science, Specialization in Machine Learning and Neural Computation Minor in Computer Science

- **Coursework:** Advanced Machine Learning, Supervised ML Algorithms, Brain-Computer Interfaces, Data Science, Neural Signal Processing, Data Structures, Probability, Neuroanatomy and Physiology, Linear Algebra
- **Honors:** UC Scholar, Provost Honors (several quarters)

Experience

Co-President *La Jolla, Ca*
UCSD Tritons Prosthetics Society *June 2025 – Present*

- Lead a multidisciplinary student team focused on prosthetic design and human-centered engineering
- Organize workshops, technical training, and collaborative projects for 50+ members across engineering and cognitive science disciplines
- Coordinate sponsorships, partnerships, and project development cycles to support student innovation in prosthetics

Research Assistant & UC Scholar *La Jolla, CA*
UCSD Cognitive Science Department *Oct 2024 – Present*

- Selected as a UC Scholar for a competitive undergraduate research program under Dr. Andrea Chiba
- Conducted research on classroom behavior and well-being using multimodal biometric, video, and audio streams
- Implemented preprocessing and signal processing pipelines for biosensor data (PPG, EDA, TEMP, ACC)
- Built large-scale workflows aligning biometric, video, and audio data with millisecond precision
- Designed and troubleshot network infrastructure for synchronized device communication (PTP, Dante)
- Produced a research paper analyzing biometric data; presented findings at the UCSD Summer Research Conference.

Data Science Intern *Amman, Jordan*
Abwaab *Aug 2024 - Dec 2024*

- Collaborated with the Learning and Data Science teams to explore predictive modeling of student performance on Abwaab's educational platform
- Contributed to early-stage feature engineering using course engagement and assessment data
- Identified key learning features and metrics based on platform usage, course interactions, and quiz performance
- Participated in team discussions around feature selection, algorithm choice, and project scope

Projects

Sarcastic Headline Generator *Code and Report*

- Trained and compared RNN, LSTM, and GRU models on satirical headlines to evaluate grammar, sarcasm, and realism
- Tools Used: Python, NLP

Comparing Supervised Learning Algorithms

- Implemented and evaluated logistic regression, KNN, and SVM across Iris, Wine, and EMG datasets using cross-validation and performance metrics
- Tools Used: Python, scikit-learn

Zebrafish Neural Activity Analysis

- Analyzed large-scale larval zebrafish brain activity using PCA and k-means to classify neural profiles and visualize region-specific dynamics
- Tools Used: Python: pandas, scikit-learn, matplotlib, NumPy

Multiscale Brain Dynamics using ICG *Code*

- Implemented and adapted an Iterative Coarse Graining model from neuroscience literature to uncover hierarchical spatial organization in zebrafish brain data
- Tools Used: Python

EEG Attention Analysis *Data, Code, and Slides*

- Designed and conducted EEG experiments, preprocessing signals and performing spectral analysis of alpha/theta bands to study attentional shifts
- Tools Used: Python, Signal Processing, EEG

Songbird Vocal Motor Analysis

- Analyzed neural spike trains to identify anticipatory activity before song initiation, using time-locked firing rate comparisons
- Tools Used: Python, Signal Processing Techniques